Train / Validation / Test Strategy & Evaluation Protocol

Smart Energy Forecasting for the CeDInt Building at UPM

1. Purpose

This document defines the evaluation strategy that will be used for the forecasting cases established in Task 5. Its objective is to ensure that all baseline and machine-learning models are trained, validated and tested under the same chronological conditions.

Since this is a forecasting problem, future observations must never be used to predict earlier periods. For this reason, the dataset will not be randomly divided. Training, validation and test periods will follow chronological order, and the final test period will remain unseen during model selection and tuning. This follows the Sprint 2 requirement to evaluate forecasting performance on later unseen periods.

2. Evaluation principles

The evaluation strategy follows the modelling decisions established in Task 5:

- hourly modelling resolution;
- 24-hour forecasting horizon;
- *energy_Wh* as the primary target;
- *W* as a secondary target;
- *I* as an exploratory secondary target;
- historical 2020–2023 and recent 2025 data treated as separate temporal regimes;
- no future information may be included in model inputs.

The same evaluation principle will be used for simple baselines and machine-learning models so that their results can be compared fairly.

3. Forecast horizons

The evaluation will consider three forecasting horizons for each target variable (energy_Wh, W and I):

- +1 hour: predict the target value for the next hourly interval (t+1). This represents the short-term forecasting case.
- +6 hours: predict the target value six hours after the forecast issue time (t+6). This represents a medium-term forecasting case.
- Next 24 hours: predict the complete hourly profile for the following day, producing 24 consecutive predictions from t+1 to t+24.

All horizons will use the same chronological train, validation and test periods to ensure a fair comparison.

For the 24-hour forecast, performance will be evaluated over the complete 24-hour prediction window. Where useful, errors may also be reported separately by lead time (h+1, h+2, …, h+24) to analyse how forecast accuracy changes as the horizon increases.

Only information available at the forecast issue time will be used. No future target values or future observations unavailable at prediction time may be included in the input features.

MAE and RMSE will be calculated separately for each target and forecasting horizon.

4. Main chronological split — Recent 2025 regime

The recent 2025 regime will be used as the main Sprint 2 modelling and evaluation case.

Before selecting the dates, hourly coverage was analysed for every available zone/series. The chosen periods provide high coverage across NE, NW, SE and SW while maintaining complete chronological separation.

Dataset	Period	Purpose
Training	19 Aug 2025 – 21 Sep 2025	Model fitting and calculation of training-based statistics
Validation	22 Sep 2025 – 28 Sep 2025	Feature, model and parameter selection
Test	29 Sep 2025 – 5 Oct 2025	Final evaluation on unseen data

The first partial day, 18 August, is excluded from the common modelling window. Data after 5 October remain outside the first evaluation protocol and may be used later for additional analysis or Sprint 3 work.

The validation and test periods each contain seven calendar days, which allows complete weekly behaviour to be evaluated and is compatible with the 24-hour forecasting horizon defined in Task 5.

5. Series eligibility and missing-data rules

A common eligibility rule is proposed to ensure that comparisons are made using sufficiently complete series.

A zone/series should contain at least **90% of the expected hourly periods in both validation and test** to be included in the main comparison.

The 90% value is a modelling-team threshold selected after reviewing the observed hourly coverage; it is not a property of the original dataset.

Missing target observations will **not** be replaced with zero. Hours where the real target value cannot be constructed reliably will be excluded from metric calculation and reported as data-quality limitations.

For example, the coverage analysis detected a small number of hours without valid *energy_Wh* in some series. These observations will remain traceable rather than being silently corrected.

This follows Task 5 rules stating that missing values must not be interpreted as zero and that series with insufficient coverage should be excluded or analysed separately.

6. Baseline forecasts

At least two simple baseline methods will be evaluated before assessing the machine-learning models. Their purpose is to determine whether additional model complexity provides real forecasting value.

Baseline 1 — Same hour previous week

For each forecasted hour, the predicted value will be the observed value from the same hour **168 hours earlier**.

For example:

Monday 10:00 forecast = value observed the previous Monday at 10:00.

This baseline represents weekly repetition in building activity.

If the corresponding value from the previous week is unavailable, the prediction will be marked unavailable rather than artificially filled.

Baseline 2 — Weekday/hour historical median

For each combination of weekday and hour, the median target value will be calculated using **training data only**.

For example, the forecast for Tuesday at 14:00 will use the median of previous Tuesdays at 14:00 contained in the training period.

Validation and test observations will never be used to calculate these medians.

Both approaches are directly aligned with the simple baseline examples identified in the Task 5 modelling strategy.

7. Evaluation metrics

MAE (Mean Absolute Error) will be used as the primary forecasting metric.

MAE provides an easily interpretable measure of the average absolute difference between predicted and observed values.

RMSE (Root Mean Squared Error) will be used as the secondary metric. RMSE gives greater importance to large prediction errors and can therefore help identify models that occasionally produce large deviations.

MAPE will not be used as the main metric because some target values may be zero or close to zero, which can make percentage errors unstable or misleading. This follows the metric recommendations defined in Task 5.

For the 24-hour forecast, results should be reported both:

- overall across the evaluation period; and

- by forecast horizon, from **h+1 to h+24**, where feasible.

This will allow the team to identify whether forecasting accuracy decreases as the prediction horizon becomes longer.

8. Fair model comparison

All candidate models and baselines must be evaluated using the **same target observations and the same chronological test period**.

Where a prediction cannot be produced by one method because the required historical information is unavailable, this must be reported. Model comparisons should use comparable valid observations whenever possible.

Model selection will be based on validation results. The test set will only be used after the model configuration has been chosen.

This is consistent with the Product Backlog requirement that candidate models use the same time-based evaluation principle and comparable metrics.

9. Leakage prevention

The following rules will be applied to prevent future-data leakage.

Calendar features such as hour, weekday, month, weekend and holiday status may be calculated because they are known at forecast time.

Lagged target variables may only use observations that occurred before the forecast issue time.

Any transformation learned from the data, such as scaling, medians or other fitted preprocessing parameters, must be fitted using training data only.

Validation and test target values must never be used to construct model inputs.

Future observed weather cannot be used unless the modelling case explicitly assumes that an equivalent weather forecast would have been available at prediction time.

These rules follow Task 5, which prohibits future targets, future-containing aggregates and transformations fitted using validation or test data.

10. Historical 2020–2023 regime

The historical 2020–2023 regime will be evaluated separately from the recent 2025 regime.

The two periods will **not** be concatenated into one continuous time series because they are separated by a long gap and have different original sampling characteristics. Task 5 explicitly requires the regimes to remain distinct unless validation evidence supports combining them.

For a secondary historical evaluation, the following chronological split is proposed after reviewing the available historical coverage:

Dataset	Period
Training	3 Apr 2020 – 30 Sep 2022
Validation	1 Oct 2022 – 31 Oct 2022
Test	1 Nov 2022 – 30 Nov 2022

SW|RACK will not participate in this common historical comparison because its available observations end in December 2021. This limitation was already identified during the final data QA. Task3_Task4_Final_QA_Summary

The historical evaluation will therefore be reported separately and will not be used as if it were continuous with the 2025 forecasting case.

11. Reproducibility

The frozen source dataset must remain unchanged.

The evaluation code will create derived modelling datasets and split indicators without overwriting the frozen input. The dataset version and relevant configuration should be recorded with each experiment.

The same split dates, metric definitions and baseline rules must be reusable by every member of the modelling team.

This maintains the distinction established in Task 5 between the frozen source dataset and derived modelling datasets.

12. Results reporting format

Target	Horizon	Model	MAE	RMSE	Regime
energy_Wh	+ 1h	Baseline	TBD	TBD	2025
energy_Wh	+ 6h	Baseline	TBD	TBD	2025
energy_Wh	24h	Baseline	TBD	TBD	2025
W	+ 1h	Baseline	TBD	TBD	2025
W	+ 6h	Baseline	TBD	TBD	2025
W	24h	Baseline	TBD	TBD	2025
I	+ 1h	Baseline	TBD	TBD	2025
I	+ 6h	Baseline	TBD	TBD	2025
I	24h	Baseline	TBD	TBD	2025